

TECHNOLOGY CONSULTING

IN THE GLOBAL COMMUNITY

Final Consulting Report

Peru SENATI

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SENATI

Executive Summary

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I. About the Organization

Founded in 1961, SENATI is the leading technical and vocational training institute in Peru. It has over 150,000 students and 64 campuses across the 13 demarcation zones in which it operates in Peru.

The organization's vision is "to lead excellence in technological and vocational education across Latin America," and its mission is

To train innovative and highly productive technical professionals .

SENATI primarily fulfills this mission by offering three-year technical degree programs, through which students receive a certification in the name of the nation in a specific area of expertise upon completion. The organization has over 5,820 employees across its campuses, 20 percent being administrative staff working under 10 departments. This consulting project was developed under the Development department with collaboration with the Finance and IT departments.

Given its large-scale operation and high number of students, SENATI relies on multiple information systems to coordinate its campuses, manage operations, and handle large volumes of data, which could still be more effectively tracked and analyzed.

II. Easily retrieve and analyze financial data on scholarship for monthly reports

Opportunity

Preparing monthly financial reports on scholarships is currently a complex and time-consuming process, taking about 28 hours from retrieving to applying the manual transformation on excel by both Finance and Development departments.

Outcome

A Power BI dashboard was developed to automate data transformations with power query and displays all tables and visuals needed for the report, reducing preparation time from 28 hours to approximately five minutes. User manuals and guidelines documentation were created to support consistent use and technical support by the IT department.

Key Risks and Recommendations

- Risks: Dashboard cannot be published online due to SENATI power BI publishing server limitations; financial data remains is retrieved in excel sheet from JD Edwards rather than the Datamart SQL server which requires new monthly data to be updated manually.

- Recommendations: Upgrade BI publishing server, move financial data to Datamart SQL server so that only a refresh is required to get updated data.

III. Increasing Data Accessibility for the Development Department: Live and Interactive Power BI

Opportunity

The existing scholarship dashboard built last year by the previous student consultant does not have up-to-date data, has inaccuracies, and lacks some key indicators, limiting its usefulness for the Development team.

Outcome

The dashboard was updated to include 2025 data, audited to correct inaccuracies and present their causes, and add new indicators. A user-friendly manual was created, and IT technical support was agreed on and recommendations were shared to them for better long term maintenance.

Key Risks and Recommendations

- Risks: Dashboard cannot be published due to SENATI Power BI publishing server limitations; requires manual path updates for Excel data once downloaded locally; changes in SQL server tables can break formulas.
- Recommendations: Upgrade BI publishing server, move historical data to Datamart SQL server, and create a merged fact table in the SQL server for improved data reliability.

IV. Recommendations

- Upgrade the BI publishing server to support map visuals for online access.
- Move all financial and historical data (2014–2018) to the Datamart SQL server to centralize data and remove dependence on Excel files and update dashboards to connect to these SQL tables
- Standardize data input formats in Apex to enhance data quality and better categorization for analytics use.
- Integrate AI features, such as a chatbot to answer user questions directly within the dashboard and an automated report generation tool to create financial reports or slide decks based on user instructions.

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SENATI

Final Consulting Report

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I. About the Organization

Organization

SENATI is the leading technical and vocational training institute in Peru. It was founded by the National Society of Industries in Peru to address the country's shortage of technical professionals and was established by law in 1961, with its first headquarters in the Los Olivos district of Lima. To date, SENATI has over 150,000 students and 64 campuses across the 13 demarcation zones in which it operates in Peru.

The organization's vision is "to lead excellence in technological and vocational education across Latin America," and its mission is

To train innovative and highly productive technical professionals .

SENATI primarily fulfills this mission by offering three-year degree programs, through which students receive national certification in a specific area of expertise upon completion.

The organization has over 5,820 employees across its campuses, with 80 percent being instructors and 20 percent administrative staff. Most administrative operations are based in the Lima office, which operates at a national level, while each campus has its own smaller local staff.

Given its large-scale operation and high number of students, SENATI relies on multiple information systems to coordinate its entities, manage operations, and handle large volumes of data, which could still be more effectively tracked and analyzed.

Facilities

SENATI's 64 campuses vary in size depending on the population density of the zones in which they are located. The central and largest campus is in Lima, which has buildings for most of the 13 schools SENATI offers, along with gardens and sports facilities. The school buildings include classrooms, clusters, labs, auditoriums, and industrial machines, depending on the field of study.

Next to the campus is an eight-level building that houses the national management and its ten departments. The levels are organized to host one to three departments, arranged in cubicles, with each cubicle equipped with a Windows desktop computer. There are also various meeting rooms, some with TVs for presentations and others without, as well as private offices for department heads.

Programs

The organization currently offers 81 careers across 13 technical schools, ranging from "Maintenance of Machinery and Industrial Plants" to "Hospitality, Tourism, and Restaurants." These careers are typically three-year degree programs in which students complete general studies, basic training in their respective fields, and dual training. Dual training combines academic learning at SENATI with practical training at one of its partner companies. Within these 81 careers, SENATI

offers more than 5,600 courses. Additionally, SENATI provides six virtual professional programs. The degrees available at each campus depend on the industrial needs of the zone in which the campus is located.

Partnerships with companies and governmental organizations are also central to SENATI's model. These partnerships enable the institute to offer scholarships to students, deliver specialized short-term training programs provided by partner organizations using their tools, offer discounted IT software licenses, and provide students with opportunities for practical training within these organizations.

Staff

At its highest level, SENATI has a national board composed of the President of SENATI and a group of entrepreneurs. In parallel, there is a National Advisory Commission that oversees academic decisions and ensures the quality of instruction at SENATI.

Below this level is the functional management, which includes ten departments located in the Lima office: Academics, Marketing, Development, Human Resources, Finances, Information Technology, Infrastructure, Legal, Commerce, and Logistics. Each department is led by a head manager who reports to the national director.

The Development (Desarrollo) Department, where this consulting project is based, is responsible for managing partnerships and scholarships, both national and international collaborations. The department is composed of eight specialists, a head manager, and an assistant.

There are 3,843 scholarship programs in total, of which 2,700 are governmental. The Development Department manages 2,850 of these scholarship programs. This project will mainly work in communication with 4 specialists: Gianinna and Karen (who have access to data from different SENATI sources), Kiko (responsible for managing PRONABEC scholarships, which are government-funded), and Giancarla (who manages all other non-governmental scholarships).

All staff use Microsoft 365 and Windows operating systems.

Technology Infrastructure

SENATI's technology infrastructure includes Ellucian Cloud, a cloud-based Enterprise Resource Planning (ERP) system specialized for higher education. It hosts most of the software and IT systems used by the organization, as well as the applications students use for their courses.

Category	Tool/Application	Description	Notes/Purpose
Ellucian cloud	Banner	Holds student information input by demarcation zones. Locating data requires knowing many keys, making data analysis use limited. Banner is	Other platforms offer views of Banner data due to access complexity.

		an SQL server.	
	Oracle Apex	Provides users drop down selections to view customized Banner tables. Many selections require parameters such as period, location or ID. Users can only view selections they're granted access to.	Implemented over 15 years ago. Categories include reports on operations, finance, HR, academics, etc.
	JD Edwards	Provides access to Banner tables. Contains financial information.	Developed in house and implemented in May 2022.
	Plus	Newer application with search functionality by ID.	Built with Javascript
	BlackBaud	Contains academic information and manages courses and grades.	Academic info from Blackbaud is input into Banner.
	PowerBI	Microsoft application for creating interactive dashboards and KPI reports. Connects to SQL views through DataMart.	Reports created by IT analysts upon request. Access granted at departmental or individual level.
	GLP Cusu	IT ticketing system for staff. Interface could be improved; FAQ may be outdated. Described as difficult and slow.	Staff often prefer using email. Large access grants reduced request volume significantly.
	Intranet	Previously functioned as a guide for technology resources.	Now outdated and redundant.
	Sinfo Administrator	Contains student data.	

Staff applications stored outside Ellucian Cloud	Siseg	Application for managing data in the preprofessional training program	Academic department exclusive
	Navajo 360	Academic department tool showing geographic employment data for students.	Academic department exclusive; other staff were unaware of its use despite expressing interest.
Student Softwares	Bibliotec, Canal Youtube, Electude, Cisco Networking Academy, SAP University Alliances, Adobe Products, Audaces, WGSN, Labor Board, and an IT Ticketing System.	Wide range of learning and support applications.	

A high-level issue with this technology infrastructure is the lack of a detailed user manual to navigate Ellucian Cloud and its associated applications. Additionally, some applications are siloed within specific departments, which can cause delays when other departments attempt to adopt them. For example, the JD Edwards system is primarily used by the Finance Department, and the IT team has communicated that licenses are no longer available when the Development (Desarrollo) Department requested access.

Technology Management and Planning

SENATI has an IT department that oversees all technology-related matters. This department is responsible for maintaining and developing the technology infrastructure, addressing the technological needs of staff and students, and providing technical support and troubleshooting for hardware and software issues. The IT department is divided into five sub-teams: Infrastructure, Project Management, Application Development, Analytics, and Tech Support.

Staff and students can report IT issues using the ticketing system CUSU. Once a ticket is submitted, the IT Tech Support team responds by either contacting the user directly or resolving the issue remotely. The request is then closed once the problem is resolved.

Although this formal support system is in place, many staff members prefer to use informal channels such as email or messaging to contact the IT team, as these methods are often perceived as faster and more convenient.

SENATI also collaborates with various technology companies hired for different aspects of management. However, the software level of accessibility with its code and data depends on the profile and responsibilities that each staff member has.

Contracted Company	Role / Responsibility	Notes
System of Contributions	Manages taxpayer data for SENATI	Access restricted
UPlanner	Analyzes student dropout risk	Uses ML algorithms, one student consultant worked on it last year
Red Logic	Supports application development	Helped implement JD Edwards; used for 3–5 years.
Evotech	Maintains databases	Provides database support; used for 3–5 years.
TeleCom	Ensures data security	Supports data protection; used for 3–5 years.
Talent Studio	Updates Power BI data	Specialized support for Power BI data management.
Carat	Develops predictive client model for Marketing	

The IT Department is responsible for all technological planning and control at SENATI, with the head of the department being primarily responsible for planning and budgeting the tools required to implement improvements within the institution.

Communication

For external communication, SENATI has a public website that provides information about the institute and all the services it offers. The Marketing and Commercial department also produces reports, publications, and market studies, and the institution maintains an active LinkedIn account for advertising purposes. Formal communication with partners is carried out through Outlook.

Internally, teams communicate using Microsoft 365 tools. Outlook is used for more formal discussions, while Teams is used for instant messaging and virtual meetings.

The Development department, along with many other departments, now follows a hybrid work system. They work online on Mondays and Fridays to avoid heavy traffic and are present in person on Tuesdays, Wednesdays, and Thursdays. This makes tools like Outlook and Teams a central part of their communication.

However, some SENATI departments can be isolated, which can cause delays and slowdowns in interdepartmental communication.

Information Management

Information is managed using database systems such as Apex and JD Edwards, although some information is also processed informally.

In the Desarrollo department, records of scholarships and the students who benefit from them can be found in Apex. However, scholarship agreements and partnership records, which are signed contracts between SENATI and other parties, are kept informally by the specialists who manage them. Each specialist has their own set of scholarships they oversee.

The Development department also has a Power BI dashboard about scholarships and student development created by a student consultant last year. However, this dashboard is not used frequently because it contains in some instances not up-to-date or partially correct information.

Financial information about scholarships and their payments is stored in JD Edwards and managed by the Finance Department. The Development team only has partial access to JD Edwards, which can cause delays in obtaining financial data. When the Development team needs financial information about their scholarships, they request it from Finance, which then provides the data in Excel sheets processed from JD Edwards about the different scholarships they manage.

The raw data from JD Edwards is complex, highly detailed, and contains many coded elements. Therefore, the Finance team performs extensive manual transformations in Excel to produce a more understandable version before sending it to the Development department. Even then, the data remains complex and granular, requiring the Development department to further transform it before using it in their reports.

Business Systems

Business systems are managed by the Accountability and Finance departments. They use information systems such as Banners to input data from bank transactions, JD Edwards to generate financial sheets, and Excel to perform financial calculations needed for their reports.

They are also responsible for providing financial information to other departments upon request.

II. Easily retrieve and analyze financial data on scholarship for monthly reports

Motivation

Problem

On average, every month, the Head of the Development department asks for financial reports on the scholarships that the department manages. Creating these financial reports can be a major hassle for the team because they have to request the information from Finance. Finance then has to perform manual transformations on excel on the data, after importing it from JD Edwards, before sending it to the Development department. After the Development department receives these detailed Excel sheets on all scholarships, they create more summarized tables using Excel. This responsibility falls mainly on Kiko, a specialist in the Development department, who manages the largest number of scholarships.

This process of making these financial reports is complex, tedious, and time-consuming for the Development and finance department. From finance retrieving the excel sheet data on JD Edwards to Kiko performing the manual transformations in Excel, this process takes on average about 28 hours to make it in a month.

Solution

Our solution is a Power BI dashboard that has all the tables and visualizations needed to make the financial reports, automating the transformations of the JD Edwards Excel sheet through Power Query. This aims to streamline the existing tedious process by automating the transformations and analyses that need to be done by the Finance and the Development department, and making them accessible now with minimal effort.

Making the process of preparing these financial reports easier is important because it is a frequent and essential task for the Development department which would help save significant amounts of time and support better decision making. There is a significant opportunity to make this process more efficient and accurate in terms of data analysis by switching from a manual process to an automated one.

Outcomes

Activities & Outputs

Activity	Output
Understand the data and the manual transformations performed monthly by Finance and the Development department	Documentation of the manual transformations performed
Review an example of the financial report to identify which information (tables and visuals) needs to be included on the dashboard	Initial list of dashboard requirements
Meet with the Development team (Kiko and others) to confirm their needs for the dashboard	Complete and revised list of financial dashboard requirements
Discuss the JD Edwards data format with Finance and agree on how data should be	Agreed process with Finance, including a SharePoint folder for monthly uploads, a

shared monthly going forward	guideline document specifying format, upload method, and deadlines; historical data from 2023 to June 2025 obtained; and a Teams group created for Finance to notify Desarrollo when data is uploaded
Automate the data transformations using Power Query	JD Edwards data transformed appropriately for dashboard visuals
Develop the first version of the dashboard in Power BI based on the requirements	First version of the dashboard with three main pages
Gather feedback from the Desarrollo team and iterate accordingly	Revised dashboard version with updated and additional visuals
Create a written user manual and a slide-based manual with only images	Written manual documenting the Power BI solution and its use, plus an image-based slide manual
Upload the dashboard folder to the cloud, including the Power BI PBIX file and supporting data folders	Dashboard folder available on SharePoint for the entire team to download and use
Conduct user testing of the guides with the Desarrollo team by presenting the manuals and assigning tasks to complete independently	Confirmation that the manuals are clear, user-friendly, and enable task completion without assistance
Meet with the IT team to explain the project and request ongoing technical support for the Desarrollo team and recommendations for long term maintenance	Agreement from the IT analytics team to provide technical support for the dashboard and a recommendation document provided to them for long term improvement of the solution

Outcomes & Indicators

❖ Power BI Dashboard

The financial dashboard was built using Power BI with Power Query used to transform the data. The dashboard contains three pages: Resumen, Pronabec, and Convenio. Resumen provides an overview of financial information for all scholarships, Pronabec contains visuals and tables about government scholarships, and Convenio focuses on company scholarships.

It was built using three data sources:

- Scholarship details (Academico.FactBecaDetallado) from the Datamart SQL Server
- JD Edwards Excel sheet data on Pronabec scholarships from 2023 provided by Ms. Gladys Cardena from Finance
- JD Edwards Excel sheet data on Convenio scholarships from 2023 provided by Mr. Jose Carranza from Finance

Power Queries were used to transform the Excel data, and now, whenever new data is added for Pronabec and Convenio, these transformations are automatically applied and the visuals are updated to include the new data.

This dashboard allows the Desarrollo team to see information about the number of scholarships, their estimated value, their actual value, and understand how the total value is distributed by zone, month, year, school period, and scholarship category.

Previously, it took about 28 hours to complete the entire process of retrieving data from JD Edwards, performing multiple manual transformations by Finance and Desarrollo, and generating the financial reports. Now, it takes about five minutes to upload the new monthly data to the dashboard folder, after which all transformations are applied automatically, and the visuals and tables needed for the financial reports are immediately available.

❖ Written Manual

The written manual is a Word document that documents the solution: what the dashboard is, what data was used, and provides instructions on how to install and set it up locally, how to update the data, and how to obtain the visuals if they need to be included in reports.

Both the Desarrollo and IT teams were able to read and fully understand how the solution works and follow all instructions.

❖ Image Manual

The image manual is a PowerPoint slide deck that provides instructions on how to install and set up the dashboard locally, update the data, and obtain visuals using only images. These are the tasks that users of the dashboard need to perform, and the image manual makes it easy for non-technical users like the Desarrollo team to follow and complete them independently.

Three members of the Desarrollo team were able to follow the instructions and complete the different manual tasks without any issues.

❖ Guidelines for Finance on Providing Data

The Finance guideline is a Word document that provides instructions based on the consensus reached with the Finance department. It specifies that Finance will upload the monthly Pronabec and Convenio data to a shared SharePoint folder by the 15th of each month. The guideline also outlines the agreed format for the JD Edwards data and what it should contain, and it includes a note about notifying the Desarrollo team through a created Teams group once the data is uploaded.

As a result, the Finance team has been able to provide the data consistently with less back and forth each month, saving approximately four hours that were previously spent on manual transformations. Now, they only need to retrieve the data from JD Edwards and upload it to the folder.

❖ IT Recommendation Guide

The IT recommendation guide is a Word document provided to the IT Analytics department, who are responsible for technical support for the dashboard. It gives recommendations on how to improve and maintain the dashboard in the long term, such as making all data available on the SQL Server and publishing the dashboard.

These recommendations have been discussed with IT, and an oral agreement was reached. If these recommendations are implemented, no further effort would be needed from the Development team to maintain the dashboard.

Risks to Sustainability

Publishing the Power BI dashboard online requires either a continuous premium account or can be published through the SENATI IT Power BI publishing server but unfortunately the latter does not support map visuals, which are included on all pages of the dashboard and are essential for the Desarrollo team. Therefore, the decision was made to keep the dashboard local rather than publishing it. While all interactive features remain available locally through the Power BI application, this approach requires users to download and set up the dashboard themselves. If it were published, it would be directly accessible via a link without setup.

Additionally, because the financial data (Pronabec and Convenio) is not available on the SQL server and is instead stored in a subfolder within the Desarrollo dashboard folder, each time someone downloads the folder and sets up the dashboard, they need to update the data folder paths in Power BI to match their local computer paths. Furthermore, when adding new monthly data, users must convert the Excel sheet to a table and place it in the appropriate data folder. Although these steps are not technically difficult, they do add extra work for maintaining and updating the dashboard.

Currently, making this financial data available on the Datamart SQL server would take about six weeks to implement which would have delayed the project too much. The data is only available as Excel sheets from JD Edwards as of now, to which Desarrollo does not have access, meaning they rely on Finance to provide the Pronabec and Convenio Excel sheets each month. Given these constraints, the current approach was the best available option.

To mitigate these challenges and risks, the IT Analytics department has agreed to assist Desarrollo with any technical issues related to the dashboard. Additionally, both a written manual and an image-based manual have been provided to Desarrollo, containing information about the dashboard and step-by-step instructions for all necessary tasks.

Another risk is that when Finance provides the JD Edwards data, they select only the needed columns from the JD Edwards interface, and for Convenio data, there is a column called “LM auxiliar (Modificado)” that is created manually. If any column used in the dashboard is missing from the Excel sheet for a given month, or if a column name is even slightly changed, it can cause errors in Power Query since data consistency across all Excel sheets is required.

To mitigate this risk, a guideline document has been shared with Finance detailing how to prepare and send the Excel data each month, along with a checklist of the essential columns and their exact names. In case of errors, Desarrollo can use this checklist to compare with the received Excel sheet to identify if the issue comes from missing or incorrect columns.

Finally, a recommendation guide has been provided to IT for long-term improvements to the dashboard such as having a publishing server that supports maps and making all data available in SQL server. When these recommendations are implemented, they will eliminate these risks.

Recommendations

The risks mentioned above would be resolved if the financial data from Finance were available directly through an SQL server, similar to how scholarship detail data is available on the SQL server, and if the dashboard could be published online and maintained directly by the IT department.

For making the Pronabec and Convenio data available through the Datamart SQL server, the IT department shared that building the ETL (Extract-Transform-Load) process would take about six weeks. This improvement is important because having the Pronabec and Convenio data as tables on the Datamart server, with all the necessary columns included, would remove the need for monthly manual data uploads and local path updates each time the dashboard is downloaded, significantly simplifying maintenance and reducing the potential for technical issues.

Publishing the dashboard online is also a key step, as it would make it more easily accessible without requiring downloads or the Power BI desktop application, and users could get updates simply by refreshing the page. For these reasons, it is recommended that the IT department update their BI publishing server to support the newer version of Power BI and Shape Map visualizations.

III. Increasing Data Accessibility for the Development Department: Live and Interactive Power BI

Motivation

Problem

The Development department handles a large volume of data on scholarships and students. To efficiently manage these scholarships, monitor their success, and secure future funding, the team needs to consistently track information and analyze performance. They also require general information on scholarships, sponsorships, donations, events, competitions, and employment to report to their partners.

Previously, obtaining this data took weeks through a tedious process. To address this, the previous student consultant last summer developed an interactive Power BI dashboard to make the data more accessible for the Development team.

However, some maintenance issues have arisen with this Power BI dashboard. It does not display data from 2025, and the team does not know how to fix it. Additionally, there have been data accuracy issues because the student consultant did not have time to cross-check and verify all the data. The team has reported that some figures, such as the total number of scholarship recipients, are visibly incorrect.

Furthermore, the team has reported limited use of the dashboard because it does not yet include some critical information they frequently need. The previous student consultant did not have time to implement these missing elements

Solution

Our solution is to audit and fix the existing dashboard, add the critical missing KPIs, and establish a more sustainable maintenance process to ensure the dashboard displays up-to-date data.

Reviving and improving the existing Power BI dashboard to make it a reliable and more functional tool for the Development department is crucial since the team consistently needs to track and report data on scholarships and students. It is important for this data to be as up-to-date as possible and accurate data is essential to avoid presenting misleading information. Furthermore, creating a more complete dashboard that provides the data they need will enable thorough and informed analysis.

Outcomes

Activities & Outputs

Activity	Output
Understand the dashboard and its data modelling by reviewing documentation and meeting with the developer and IT Analytics department	Dashboard data modelling understood with ideas written on how to improve it
Audit the dashboard to identify incorrect information and its causes	Documentation created outlining the causes of incorrect data and proposed solutions
Updated dashboard to include up to data and Fix incorrect information on it	Dashboard edited to include 2025 data and some of the inaccurate data corrected
Add a new page and features to the dashboard with new KPIs	New page added showing students' high school origins; a year filter and improved map visuals added to all pages
Create an image-based manual to help Desarrollo use and maintain the dashboard	Manual with slides explaining, using only images, how to download the dashboard, update data folder paths after downloading, and maintain it
Upload the dashboard folder to the cloud with the Power BI PBIX file and supporting data folders	Updated dashboard folder uploaded to SharePoint for the whole team to download and use
Meet with the IT team to explain the project and request maintenance support if the Desarrollo team faces technical issues	Agreement from the IT analytics team to provide technical support for the dashboard and a recommendation document provided to them for long term improvement of the solution

Outcomes & Indicators

❖ Updated Power BI Dashboard

The dashboard built last year has now been updated to include data from 2025, add a year dropdown filter on all pages, and include a new page displaying data on students' high school origins.

These updates improve the reliability of the dashboard data and add eight new indicators to the new page.

❖ Dashboard Auditing Document

This is a Word document containing the findings from the dashboard audit, including identified data inaccuracies and their causes, actions taken to address them, and recommendations for improving data reliability.

❖ Image Manual for the Dashboard

The image manual is a PowerPoint slide deck that provides instructions on how to install and set up the dashboard locally and how to update the data. It was created to prevent the maintenance issues that Desarrollo faced after the previous developer left.

❖ IT Recommendation Guide

The IT recommendation guide is a Word document provided to the IT Analytics department, who are responsible for technical support for the dashboard. It contains recommendations on long-term improvements related to maintenance. It is the same document as the Financial Dashboard IT recommendation guide, with specific recommendations for each dashboard included within the document.

These recommendations have been discussed with IT, and an oral agreement was reached that, if implemented, these actions would eliminate the need for ongoing maintenance efforts from the Development team.

Risks to Sustainability

For the same reasons outlined regarding the financial dashboard not being published, this dashboard also could not be published. While this does not reduce its functionality, it means that the dashboard must be downloaded and opened with the Power BI application on a laptop. Additionally, if changes are made to the dashboard, it will need to be downloaded again to access the updated version.

Furthermore, since the data sources from 2014 to 2018 are Excel sheets, the file paths for these sheets need to be updated to their local paths after downloading the dashboard. This update only needs to be done once, as no monthly data updates are required for these years. From 2019 onwards, data is retrieved directly from the Datamart SQL server and only requires a refresh. However, updating the file paths can be a tedious process due to the large number of Excel files involved.

After the dashboard was built last year, the Development team faced difficulties maintaining it because they did not know how to retrieve updated data from the SQL server. Additionally, as explained in the Audit document, some columns that were present in the SQL server tables were later removed by IT, causing errors in formulas in Power Query since those columns were mentioned in them.

To mitigate these maintenance risks, the IT Analytics department has agreed to assist Desarrollo with any technical issues related to the dashboard and to ensure the dashboard is updated accordingly if changes are made to the SQL server tables. In addition, an image-based manual has been provided to Desarrollo, explaining in simple terms how to download the dashboard, update file paths, and refresh the data to obtain the latest information.

Recommendations

We recommend, as with the financial dashboard, that the BI publishing server be updated to support map visualizations so this dashboard can be published online. Additionally, it would be more efficient to have the data from 2014–2018 also available in the Datamart SQL server so that all data can be retrieved directly from the SQL server without relying on Excel sheets.

We also advise merging the five main fact tables that currently each have a granularity of one student per period and making it available as one table in the Datamart SQL server. Having these tables merged would improve the reliability of the data displayed. Currently, the connections between fact tables are not robust, and merging them within Power Query was not possible because the queries became so large they resulted in infinite loading times.

If all data is moved to the Datamart SQL server for this dashboard, as well as for the financial dashboard, both dashboards could be combined into a single, centralized dashboard, making it easier for the Development team to access all analytics related to their operations in one place.

IV. Recommendations

Improving Data Infrastructure and Dashboard Sustainability

Overall, it falls under IT's responsibility to update their BI publishing server to support map visualizations so that both dashboards can be published and accessed directly on the web. Additionally, IT should make the financial data and the data sources from 2014–2018 available in the Datamart SQL server so that all data can be queried from a single location.

Once these changes are made, the dashboards will need to be updated to replace the Excel sheet data sources with their respective tables in the SQL server and to update the related measures accordingly. Ideally, the tables in the SQL server will have a similar format to the current Excel sheets, so that only duplicating the Power Query transformations and updating the table names in the measures will be necessary, rather than having to change the underlying logic.

Finally, we recommend implementing better standardization of data input across all tables used. Adding dropdown options for data entry would improve categorization consistency and enhance overall data quality.

AI-Powered Chatbot and Automated Report Generation

Another recommendation is to implement an AI-powered chatbot within the dashboard, using the OpenAI API or another AI model. This would allow users to ask any questions they have about the dashboard or obtain insights interactively while using it.

Additionally, while the Development department currently creates financial reports manually using tables and visuals from the financial dashboard, it would be beneficial to implement a feature enabling automated report or slide deck generation. An AI system could generate these reports using the tables and visuals from the dashboard, based on user-provided instructions.

If the two dashboards are eventually integrated into a single centralized dashboard, this AI feature could serve as a comprehensive tool to ask questions and generate reports about any data contained within the unified system.

About the Consultant

Fatou Gueye is a senior in Information Systems concentrating in Data Science with minors in Human Computer Interaction and Business Administration at Carnegie Mellon University in Qatar. She was taking part in the Technology Consulting in the Global Community internship over the summer and will be returning back to school to finish her bachelor's degree. She is interested in going to graduate school and a career at the intersection of technology and business.